

25. Port Output Enable 3 (POE3a)

This MCU incorporates a port output enable 3 (POE3a) which can be used to, under various conditions, disable output signals for the MTU. Every output signal is put in the high-impedance state when the output is disabled.

In this section, “PCLK” is used to refer to PCLKB.

25.1 Overview

Table 25.1 lists the specifications of the POE3, and Figure 25.1 shows a block diagram of the POE3.

Table 25.1 POE3 Specifications

Item	Description														
Pin status while output is disabled	High-impedance														
Target pins for switching to high-impedance state	- MTU output pins - MTU0 pins (MTIOC0A, MTIOC0B, MTIOC0C, MTIOC0D) - MTU3 pins (MTIOC3B, MTIOC3D) - MTU4 pins (MTIOC4A, MTIOC4B, MTIOC4C, MTIOC4D) - MTU6 pins (MTIOC6B, MTIOC6D) - MTU7 pins (MTIOC7A, MTIOC7B, MTIOC7C, MTIOC7D)														
Generating conditions of request for switching to high-impedance state	- Input signal detection: Detection of the POE0#, POE4#, POE8#, POE10#, and POE11# signal level. - Simultaneous conduction between output pins: A match (simultaneous conduction) between the output signal levels at the active level over one or more cycles on the following combination of pins <table border="1" data-bbox="437 981 922 1256"> <thead> <tr> <th></th><th>MTU Complementary PWM Output Pins</th></tr> </thead> <tbody> <tr> <td>1</td><td>MTIOC3B and MTIOC3D</td></tr> <tr> <td>2</td><td>MTIOC4A and MTIOC4C</td></tr> <tr> <td>3</td><td>MTIOC4B and MTIOC4D</td></tr> <tr> <td>4</td><td>MTIOC6B and MTIOC6D</td></tr> <tr> <td>5</td><td>MTIOC7A and MTIOC7C</td></tr> <tr> <td>6</td><td>MTIOC7B and MTIOC7D</td></tr> </tbody> </table> - The SPOER register setting - Detection that the main clock oscillator had stopped oscillating		MTU Complementary PWM Output Pins	1	MTIOC3B and MTIOC3D	2	MTIOC4A and MTIOC4C	3	MTIOC4B and MTIOC4D	4	MTIOC6B and MTIOC6D	5	MTIOC7A and MTIOC7C	6	MTIOC7B and MTIOC7D
	MTU Complementary PWM Output Pins														
1	MTIOC3B and MTIOC3D														
2	MTIOC4A and MTIOC4C														
3	MTIOC4B and MTIOC4D														
4	MTIOC6B and MTIOC6D														
5	MTIOC7A and MTIOC7C														
6	MTIOC7B and MTIOC7D														
Function	- Each of the POE0#, POE4#, POE8#, POE10#, and POE11# input pins can be set for falling edge, PCLK/8 × 16, PCLK/16 × 16, or PCLK/128 × 16 low-level sampling. - The outputs of the target pins can be in the high-impedance state by falling-edge or low-level sampling of the POE0#, POE4#, POE8#, POE10#, or POE11# pin. - The outputs of the target pins can be in the high-impedance state when oscillation stop is detected by the oscillation stop detection function of the clock generator. - The MTU complementary PWM outputs can be in the high-impedance state when output levels of the MTU complementary PWM output pins are compared and simultaneous active-level output continues for one cycle or more. - The outputs of the target pins can be in the high-impedance state by modifying the settings of the POE3 registers. - Interrupts can be generated by input-level sampling or output-level comparison results.														

The POE3 has input-level detection circuits, pin selection circuits, output-level comparison circuits, and a high-impedance request/interrupt request generating circuit as shown in Figure 25.1.

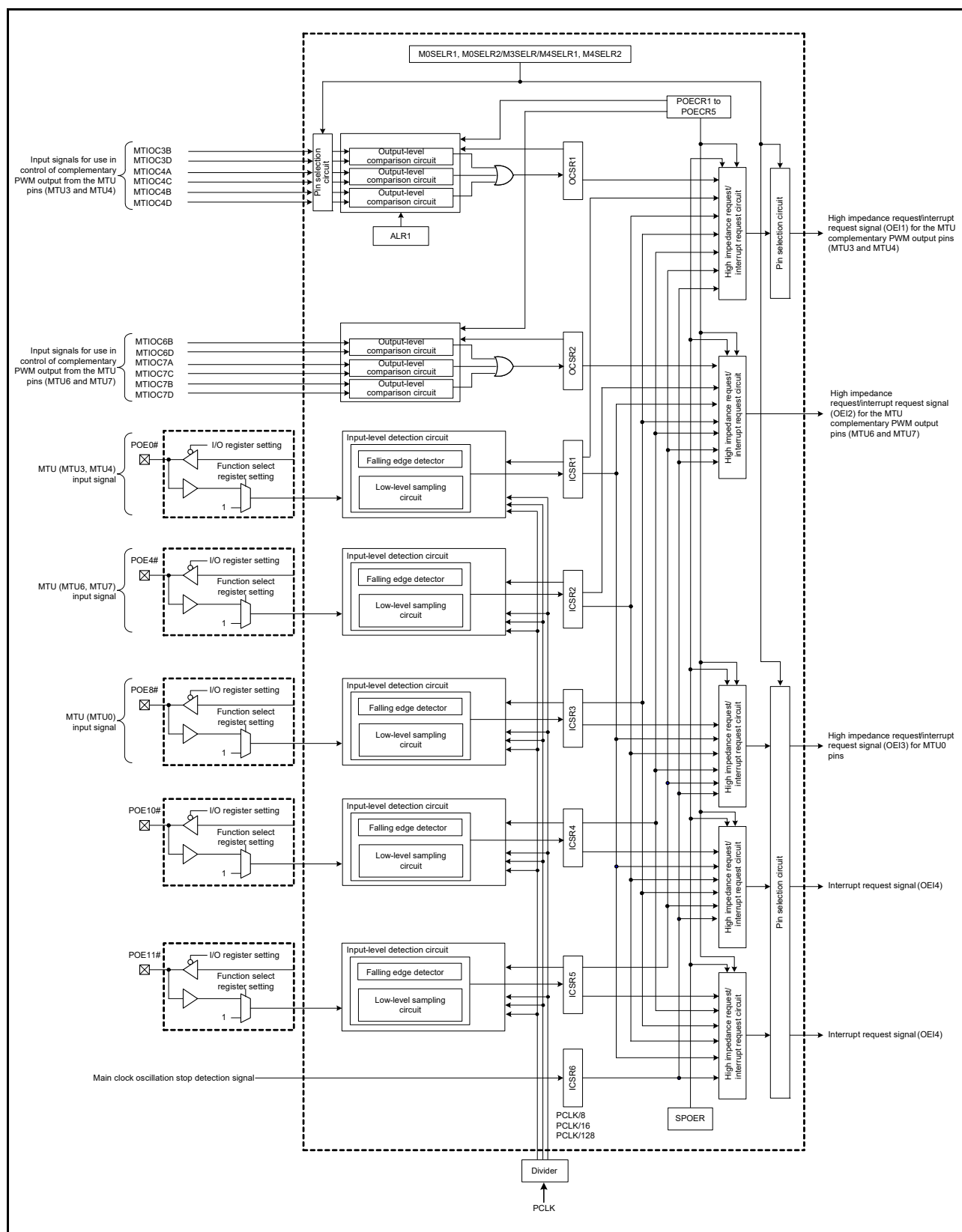


Figure 25.1 POE3 Block Diagram

Table 25.2 shows I/O pins to be used by the POE3.

Table 25.2 POE3 I/O Pins

Pin Name	I/O	Description
POE0#	Input	Request signal to put the outputs of the MTU complementary PWM output pins (MTU3, MTU4 pins) in the high-impedance state, and is also capable of controlling the other target pins by register settings.
POE4#	Input	Request signal to put the output of the MTU complementary PWM output pins (MTU6, MTU7 pins) in the high-impedance state, and is also capable of controlling the other target pins by register settings.
POE8#	Input	Request signal to put the output of the MTU0 pins in the high-impedance state, and is also capable of controlling the other target pins by register settings.
POE10#	Input	Is capable of controlling every target pins by register settings.
POE11#	Input	Is capable of controlling every target pins by register settings.

Table 25.3 shows output-level comparisons with pin combinations.

Table 25.3 Pin Combinations

Pin Combination	I/O	Description
MTIOC3B and MTIOC3D	Output	The MTU complementary PWM output pins (MTU3 and MTU4 pins) set in the M3SELR, M4SELR1, and M4SELR2 registers are in the high-impedance state when two pins of the set simultaneously output the active level (low level when the MTU.TOCR1A.OLSP bit is 0 or high level when the OLSP bit is 1 while the OLSEN bit in the ALR1 register is 0 and the MTU.TOCR1A.TOCS bit is 0, low level when the OLS1P, OLS1N, OLS2P, OLS2N, OLS3P, and OLS3N bits in the MTU.TOCR2A register are 0 or high level when these bits are 1 while the OLSEN bit in the ALR1 register is 0 and the MTU.TOCR1A.TOCS bit is 1, or low level when the OLSG0A, OLSG0B, OLSG1A, OLSG1B, OLSG2A, and OLSG2B bits in the ALR1 register are 0 and high level when these bits are 1 while the OLSEN bit in the ALR1 register is 1) for at least one cycle of the peripheral module clock (PCLK). Pin combinations for output comparison can be selected by registers of POE3.
MTIOC4A and MTIOC4C	Output	
MTIOC4B and MTIOC4D	Output	
MTIOC6B and MTIOC6D	Output	The MTU complementary PWM output pins (MTU6 and MTU7 pins) are in the high-impedance state when two pins of the set simultaneously output the active level (low level when the MTU.TOCR1B.OLSP bit is 0 or high level when the OLSP bit is 1 while the MTU.TOCR1B.TOCS bit is 0, or low level when the OLS1P, OLS1N, OLS2P, OLS2N, OLS3P, and OLS3N bits in the MTU.TOCR2B register are 0 or high level when these bits are 1 while the MTU.TOCR1B.TOCS bit is 1) for at least one cycle of the peripheral module clock (PCLK). Pin combinations for output comparison can be selected by registers of POE3.
MTIOC7A and MTIOC7C	Output	
MTIOC7B and MTIOC7D	Output	

25.2 Register Descriptions

The POE3 registers are initialized by a reset.

25.2.1 Input Level Control/Status Register 1 (ICSR1)

Address(es): POE3.ICSR1 0008 C4C0h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	POE0F	—	—	—	PIE1	—	—	—	—	—	—	POE0M[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	POE0M[1:0]	POE0 Mode Select	b1 b0 0 0: Accepts a request on the falling edge of POE0# pin input. 0 1: Accepts a request when POE0# pin input has been sampled 16 times at PCLK/8 clock pulses and all are low level. 1 0: Accepts a request when POE0# pin input has been sampled 16 times at PCLK/16 clock pulses and all are low level. 1 1: Accepts a request when POE0# pin input has been sampled 16 times at PCLK/128 clock pulses and all are low level.	R/W*1
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	PIE1	Port Interrupt Enable 1	0: Interrupt requests disabled 1: Interrupt requests enabled	R/W
b11 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	POE0F	POE0 Flag	0: Indicates that a high-impedance request has not been input to the POE0# pin. 1: Indicates that a high-impedance request has been input to the POE0# pin.	R/(W) *2
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

Note 2. Writing 0 to this bit after reading it as 1 clears the flag and is the only allowed way.

The ICSR1 register selects the POE0# pin input modes, controls the enable/disable of interrupts, and indicates status.

POE0M[1:0] Bits (POE0 Mode Select)

These bits select the input mode of the POE0# pin.

PIE1 Bit (Port Interrupt Enable 1)

This bit enables or disables interrupt requests when the POE0F flag is set to 1.

POE0F Flag (POE0 Flag)

This flag indicates that a high-impedance request has been input to the POE0# pin.

[Setting condition]

- When the input set by the POE0M[1:0] bits occurs at the POE0# pin

[Clearing condition]

- By writing 0 to the POE0F flag after reading POE0F = 1
When low-level sampling is set by the POE0M[1:0] bits, the high level needs to be input to the POE0# pin to write 0 to this flag.
For details, refer to section 25.3.7, Recover from High-Impedance State.

25.2.2 Input Level Control/Status Register 2 (ICSR2)

Address(es): POE3.ICSR2 0008 C4C4h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	POE4F	—	—	—	PIE2	—	—	—	—	—	—	POE4M[1:0]	—
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	POE4M[1:0]	POE4 Mode Select	b1 b0 0 0: Accepts a request on the falling edge of POE4# pin input. 0 1: Accepts a request when POE4# pin input has been sampled 16 times at PCLK/8 clock pulses and all are low level. 1 0: Accepts a request when POE4# pin input has been sampled 16 times at PCLK/16 clock pulses and all are low level. 1 1: Accepts a request when POE4# pin input has been sampled 16 times at PCLK/128 clock pulses and all are low level.	R/W*1
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	PIE2	Port Interrupt Enable 2	0: Interrupt requests disabled 1: Interrupt requests enabled	R/W
b11 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	POE4F	POE4 Flag	0: Indicates that a high-impedance request has not been input to the POE4# pin. 1: Indicates that a high-impedance request has been input to the POE4# pin.	R/(W) *2
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

Note 2. Writing 0 to this bit after reading it as 1 clears the flag and is the only allowed way.

The ICSR2 register selects the POE4# pin input mode, controls the enable/disable of interrupts, and indicates status.

POE4M[1:0] Bits (POE4 Mode Select)

These bits select the input mode of the POE4# pin.

PIE2 Bit (Port Interrupt Enable 2)

This bit enables or disables interrupt requests when the POE4F flag is set to 1.

POE4F Flag (POE4 Flag)

This flag indicates that a high-impedance request has been input to the POE4# pin.

[Setting condition]

- When the input set by POE4M[1:0] occurs at the POE4# pin

[Clearing condition]

- By writing 0 to POE4F after reading POE4F = 1
When low-level sampling is set by the POE4M[1:0] bits, the high level needs to be input to the POE4# pin to write 0 to this flag.
For details, refer to section 25.3.7, Recover from High-Impedance State.

25.2.3 Input Level Control/Status Register 3 (ICSR3)

Address(es): POE3.ICSR3 0008 C4C8h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	POE8F	—	—	POE8E	PIE3	—	—	—	—	—	—	POE8M[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	POE8M[1:0]	POE8 Mode Select	b1 b0 0 0: Accepts a request on the falling edge of POE8# pin input. 0 1: Accepts a request when POE8# pin input has been sampled 16 times at PCLK/8 clock pulses and all are low level. 1 0: Accepts a request when POE8# pin input has been sampled 16 times at PCLK/16 clock pulses and all are low level. 1 1: Accepts a request when POE8# pin input has been sampled 16 times at PCLK/128 clock pulses and all are low level.	R/W*1
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	PIE3	Port Interrupt Enable 3	0: Interrupt requests disabled 1: Interrupt requests enabled	R/W
b9	POE8E	POE8 High-Impedance Enable	0: Does not put the output in the high-impedance state by POE8# signal. 1: Put the output in the high-impedance state by POE8# signal.	R/W*1
b11, b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	POE8F	POE8 Flag	0: Indicates that a high-impedance request has not been input to the POE8# pin. 1: Indicates that a high-impedance request has been input to the POE8# pin.	R/(W) *2
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

Note 2. Writing 0 to this bit after reading it as 1 clears the flag and is the only allowed way.

The ICSR3 register selects the POE8# pin input mode, controls the enable/disable of interrupts, and indicates status.

POE8M[1:0] Bits (POE8 Mode Select)

These bits select the input mode of the POE8# pin.

PIE3 Bit (Port Interrupt Enable 3)

This bit enables or disables interrupt requests when the POE8F flag is set to 1.

POE8E Bit (POE8 High-Impedance Enable)

This bit specifies whether to put the output of the corresponding pin in the high-impedance state when the POE8F flag is set to 1.

POE8F Flag (POE8 Flag)

This flag indicates that a high-impedance request has been input to the POE8# pin.

[Setting condition]

- When the input set by the POE8M[1:0] bits occurs at the POE8# pin

[Clearing condition]

- By writing 0 to the POE8F flag after reading POE8F = 1
When low-level sampling is set by the POE8M[1:0] bits, the high level needs to be input to the POE8# pin to write 0 to this flag.

For details, refer to section 25.3.7, Recover from High-Impedance State.

25.2.4 Input Level Control/Status Register 4 (ICSR4)

Address(es): POE3.ICSR4 0008 C4D6h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	POE10F	—	—	POE10E	PIE4	—	—	—	—	—	—	POE10M[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	POE10M[1:0]	POE10 Mode Select	b1 b0 0 0: Accepts a request on the falling edge of POE10# pin input. 0 1: Accepts a request when POE10# pin input has been sampled 16 times at PCLK/8 clock pulses and all are low level. 1 0: Accepts a request when POE10# pin input has been sampled 16 times at PCLK/16 clock pulses and all are low level. 1 1: Accepts a request when POE10# pin input has been sampled 16 times at PCLK/128 clock pulses and all are low level.	R/W*1
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	PIE4	Port Interrupt Enable 4	0: Interrupt requests disabled 1: Interrupt requests enabled	R/W
b9	POE10E	POE10 High-Impedance Enable	0: Does not put the output in the high-impedance state by POE10# signal. 1: Put the output in the high-impedance state by POE10# signal.	R/W*1
b11, b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	POE10F	POE10 Flag	0: Indicates that a high-impedance request has not been input to the POE10# pin. 1: Indicates that a high-impedance request has been input to the POE10# pin.	R/(W)*2
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

Note 2. Writing 0 to this bit after reading it as 1 clears the flag and is the only allowed way.

The ICSR4 register selects the POE10# pin input mode, controls the enable/disable of interrupts, and indicates status.

POE10M[1:0] Bits (POE10 Mode Select)

These bits select the input mode of the POE10# pin.

PIE4 Bit (Port Interrupt Enable 4)

This bit enables or disables interrupt requests when the POE10F flag is set to 1.

POE10E Bit (POE10 High-Impedance Enable)

This bit specifies whether to put the output of the corresponding pin in the high-impedance state when the POE10F flag is set to 1.

POE10F Flag (POE10 Flag)

This flag indicates that a high-impedance request has been input to the POE10# pin.

[Setting condition]

- When the input set by the POE10M[1:0] bits occurs at the POE10# pin

[Clearing condition]

- By writing 0 to the POE10F flag after reading POE10F = 1

When low-level sampling is set by the POE10M[1:0] bits, the high level needs to be input to the POE10# pin to write 0 to this flag.

For details, refer to section 25.3.7, Recover from High-Impedance State.

25.2.5 Input Level Control/Status Register 5 (ICSR5)

Address(es): POE3.ICSR5 0008 C4D8h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	POE11F	—	—	POE11E	PIE5	—	—	—	—	—	—	POE11M[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	POE11M[1:0]	POE11 Mode Select	b1 b0 0 0: Accepts a request on the falling edge of POE11# pin input. 0 1: Accepts a request when POE11# pin input has been sampled 16 times at PCLK/8 clock pulses and all are low level. 1 0: Accepts a request when POE11# pin input has been sampled 16 times at PCLK/16 clock pulses and all are low level. 1 1: Accepts a request when POE11# pin input has been sampled 16 times at PCLK/128 clock pulses and all are low level.	R/W*1
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	PIE5	Port Interrupt Enable 5	0: Interrupt requests disabled 1: Interrupt requests enabled	R/W
b9	POE11E	POE11 High-Impedance Enable	0: Does not put the output in the high-impedance state by POE11# signal. 1: Put the output in the high-impedance state by POE11# signal.	R/W*1
b11, b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	POE11F	POE11 Flag	0: Indicates that a high-impedance request has not been input to the POE11# pin. 1: Indicates that a high-impedance request has been input to the POE11# pin.	R/(W) *2
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

Note 2. Writing 0 to this bit after reading it as 1 clears the flag and is the only allowed way.

The ICSR5 register selects the POE11# pin input mode, controls the enable/disable of interrupts, and indicates status.

POE11M[1:0] Bits (POE11 Mode Select)

These bits select the input mode of the POE11# pin.

PIE5 Bit (Port Interrupt Enable 5)

This bit enables or disables interrupt requests when the POE11F flag is set to 1.

POE11E Bit (POE11 High-Impedance Enable)

This bit specifies whether to put the output of the corresponding pin in the high-impedance state when the POE11F flag is set to 1.

POE11F Flag (POE11 Flag)

This flag indicates that a high-impedance request has been input to the POE11# pin.

[Setting condition]

- When the input set by the POE11M[1:0] bits occurs at the POE11# pin

[Clearing condition]

- By writing 0 to the POE11F flag after reading POE11F = 1
When low-level sampling is set by the POE11M[1:0] bits, the high level needs to be input to the POE11# pin to write 0 to this flag.
For details, refer to section 25.3.7, Recover from High-Impedance State.

25.2.6 Input Level Control/Status Register 6 (ICSR6)

Address(es): POE3.ICSR6 0008 C4DCh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	OSTST F	—	—	OSTST E	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b8 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b9	OSTSTE	Oscillation Stop High-Impedance Enable	0: Does not put the output in the high-impedance state when the oscillation stop is detected. 1: Put the output in the high-impedance state when the oscillation stop is detected.	R/W ^{*1}
b11, b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	OSTSTF	Oscillation Stop Detection Flag	0: Indicates that a high-impedance request by oscillation stop has not been generated. 1: Indicates that a high-impedance request by oscillation stop has been generated.	R/W ^{*2}
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

Note 2. Writing 0 to this bit after reading it as 1 clears the flag and is the only allowed way.

The ICSR6 register controls the oscillation stop high-impedance and indicates status.

OSTSTE Bit (Oscillation Stop High-Impedance Enable)

This bit specifies whether to put the output of the target pin in the high-impedance state when oscillation stop is detected.

OSTSTF Flag (Oscillation Stop Detection Flag)

This flag indicates that a high-impedance request by the oscillation stop has been generated.

When the main clock oscillation stops, this flag is set to 1. To clear this flag, wait for at least 10 cycles of PCLK after this flag becomes 1 and write 0 to this flag while the OSTDSR.OSTDF flag is 0. Writing 0 to this flag while the OSTDSR.OSTDF flag is 1 cannot clear this flag. After clearing this flag, confirm that the flag has actually been modified to 0.

[Setting condition]

- When oscillation stop is detected

[Clearing condition]

- By writing 0 to the OSTSTF flag after reading OSTSTF = 1

25.2.7 Output Level Control/Status Register 1 (OCSR1)

Address(es): POE3.OCSR1 0008 C4C2h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	OSF1	—	—	—	—	—	OCE1	OIE1	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	OIE1	Simultaneous Conduction Interrupt Enable 1	0: Interrupt requests disabled 1: Interrupt requests enabled	R/W
b9	OCE1	Simultaneous Conduction High-Impedance Enable 1	0: Does not put the outputs in the high-impedance state when they simultaneously go to an active level. 1: Put the outputs in the high-impedance state when they simultaneously go to an active level.	R/W*1
b14 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15	OSF1	Simultaneous Conduction Flag 1	0: Indicates that outputs have not simultaneously become an active level. 1: Indicates that outputs have simultaneously become an active level.	R/(W) *2

Note 1. Can be modified only once after a reset.

Note 2. Writing 0 to this bit after reading it as 1 clears the flag and is the only allowed way.

The OCSR1 register controls the enable/disable of output-level comparison and interrupts, and indicates status.

OIE1 Bit (Simultaneous Conduction Interrupt Enable 1)

This bit enables or disables interrupt requests when the OSF1 flag is set to 1.

OCE1 Bit (Simultaneous Conduction High-Impedance Enable 1)

This bit specifies whether to put the output of the target pin in the high-impedance state when the OSF1 flag is set to 1.

OSF1 Flag (Simultaneous Conduction Flag 1)

This flag indicates that at least one of the three pairs of two-phase output pins for MTU complementary PWM output (MTU3 and MTU4) has simultaneously become at the active level. If the high-impedance control for the corresponding pins is not enabled, this flag does not become 1.

For setting the active level, refer to section 25.2.9, Active Level Setting Register 1 (ALR1).

[Setting condition]

- When the MTIOC3B and MTIOC3D pins simultaneously go to the active level*1 for at least one cycle of PCLK while the POE3.MTU3BDZE bit is 1.
- When the MTIOC4A and MTIOC4C pins simultaneously go to the active level*1 for at least one cycle of PCLK while the POE3.MTU4ACZE bit is 1.
- When the MTIOC4B and MTIOC4D pins simultaneously go to the active level*1 for at least one cycle of PCLK while the POE3.MTU4BDZE bit is 1.

Note 1. The setting condition is judged only by the level of the pin regardless the setting of the MPC.PmnPFS register.

[Clearing condition]

- By writing 0 to the OSF1 flag after reading OSF1 = 1

To write 0 to this flag, the inactive level needs to be output from MTU complementary PWM output pins. For details,

refer to section 25.3.7, Recover from High-Impedance State.

25.2.8 Output Level Control/Status Register 2 (OCSR2)

Address(es): POE3.OCSR2 0008 C4C6h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	OSF2	—	—	—	—	—	OCE2	OIE2	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	OIE2	Simultaneous Conduction Interrupt Enable 2	0: Interrupt requests disabled 1: Interrupt requests enabled	R/W
b9	OCE2	Simultaneous Conduction High-Impedance Enable 2	0: Does not put the outputs in the high-impedance state when they simultaneously go to an active level. 1: Put the outputs in the high-impedance state when they simultaneously go to an active level.	R/W*1
b14 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15	OSF2	Simultaneous Conduction Flag 2	0: Indicates that outputs have not simultaneously become an active level. 1: Indicates that outputs have simultaneously become an active level.	R/(W) *2

Note 1. Can be modified only once after a reset.

Note 2. Writing 0 to this bit after reading it as 1 clears the flag and is the only allowed way.

The OCSR2 register controls the enable/disable of output-level comparison and interrupts, and indicates status.

OIE2 Bit (Simultaneous Conduction Interrupt Enable 2)

This bit enables or disables interrupt requests when the OSF2 flag is set to 1.

OCE2 Bit (Simultaneous Conduction High-Impedance Enable 2)

This bit specifies whether to put the output of the target pin in the high-impedance state when the OSF2 flag is set to 1.

OSF2 Flag (Simultaneous Conduction Flag 2)

This flag indicates that at least one of the three pairs of two-phase output pins for MTU complementary PWM output (MTU6 and MTU7) has simultaneously become an active level. If the high-impedance control for the corresponding pins is not enabled, this flag does not become 1.

For setting the active level, refer to section 24, Multi-Function Timer Pulse Unit 3 (MTU3a).

[Setting condition]

- When the MTIOC6B and MTIOC6D pins simultaneously go to the active level*1 for at least one cycle of PCLK while the POE3.MTU6BDZE bit is 1.
- When the MTIOC7A and MTIOC7C pins simultaneously go to the active level*1 for at least one cycle of PCLK while the POE3.MTU7ACZE bit is 1.
- When the MTIOC7B and MTIOC7D pins simultaneously go to the active level*1 for at least one cycle of PCLK while the POE3.MTU7BDZE bit is 1.

Note 1. The setting condition is judged only by the level of the pin regardless the setting of the MPC.PmnPFS register.

[Clearing condition]

- By writing 0 to the OSF2 flag after reading OSF2 = 1

To write 0 to this flag, the inactive level needs to be output from MTU complementary PWM output pins. For details, refer to section 25.3.7, Recover from High-Impedance State.

25.2.9 Active Level Setting Register 1 (ALR1)

Address(es): POE3.AL1 0008 C4DAh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	OLSEN	—	OLSG2B	OLSG2A	OLSG1B	OLSG1A	OLSG0B	OLSG0A
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	OLSG0A	MTIOC3B Pin Active Level Setting	0: Active low 1: Active high	R/W*1
b1	OLSG0B	MTIOC3D Pin Active Level Setting	0: Active low 1: Active high	R/W*1
b2	OLSG1A	MTIOC4A Pin Active Level Setting	0: Active low 1: Active high	R/W*1
b3	OLSG1B	MTIOC4C Pin Active Level Setting	0: Active low 1: Active high	R/W*1
b4	OLSG2A	MTIOC4B Pin Active Level Setting	0: Active low 1: Active high	R/W*1
b5	OLSG2B	MTIOC4D Pin Active Level Setting	0: Active low 1: Active high	R/W*1
b6	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7	OLSEN	Active Level Setting Enable	0: Disabled 1: Enabled	R/W*1
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

The ALR1 register specifies the active levels of the MTU outputs for detection of simultaneous conduction of those outputs as reflected in the OCSR1 register.

OLSG0A Bit (MTIOC3B Pin Active Level Setting)

This bit sets the active level of the MTIOC3B output. Specifically, setting the OLSG0A bit to 0 sets the low level and to 1 sets the high level as the active level for detection of simultaneous conduction.

OLSG0B Bit (MTIOC3D Pin Active Level Setting)

This bit sets the active level of the MTIOC3D output. Specifically, setting the OLSG0B bit to 0 sets the low level and to 1 sets the high level as the active level for detection of simultaneous conduction.

OLSG1A Bit (MTIOC4A Pin Active Level Setting)

This bit sets the active level of the MTIOC4A output. Specifically, setting the OLSG1A bit to 0 sets the low level and to 1 sets the high level as the active level for detection of simultaneous conduction.

OLSG1B Bit (MTIOC4C Pin Active Level Setting)

This bit sets the active level of the MTIOC4C output. Specifically, setting the OLSG1B bit to 0 sets the low level and to 1 sets the high level as the active level for detection of simultaneous conduction.

OLSG2A Bit (MTIOC4B Pin Active Level Setting)

This bit sets the active level of the MTIOC4B output. Specifically, setting the OLSG2A bit to 0 sets the low level and to 1 sets the high level as the active level for detection of simultaneous conduction.

OLSG2B Bit (MTIOC4D Pin Active Level Setting)

This bit sets the active level of the MTIOC4D output. Specifically, setting the OLSG2B bit to 0 sets the low level and to 1 sets the high level as the active level for detection of simultaneous conduction.

OLSEN Bit (Active Level Setting Enable)

This bit enables or disables of the active-level settings in the OLSGnm bits ($n = 0$ to 2 ; $m = A, B$). Clearing the OLSEN bit to 0 disables the OLSGnm bits, in which case the active levels of the MTU output are determined by the MTU.TOCR1j and MTU.TOCR2j registers ($j = A, B$). Setting the OLSEN bit to 1 enables the OLSGnm bits, in which case the active levels of the MTU output are as selected by the OLSGnm bits in this register.

25.2.10 Software Port Output Enable Register (SPOER)

Address(es): POE3.SPOER 0008 C4CAh

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	MTUCH0HIZ	MTUCH67HIZ	MTUCH34HIZ
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MTUCH34HIZ	MTU3 and MTU4 Pin High-Impedance Enable	0: Does not put the outputs in the high-impedance state. 1: Put the outputs in the high-impedance state.	R/W
b1	MTUCH67HIZ	MTU6 and MTU7 Pin High-Impedance Enable	0: Does not put the output in the high-impedance state. 1: Put the output in the high-impedance state.	R/W
b2	MTUCH0HIZ	MTU0 Pin High-Impedance Enable	0: Does not put the outputs in the high-impedance state. 1: Put the outputs in the high-impedance state.	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The SPOER register is used to put the outputs of the corresponding pins in the high-impedance state.

MTUCH34HIZ Bit (MTU3 and MTU4 Pin High-Impedance Enable)

This bit specifies whether to put the outputs of the MTU complementary PWM output pins (MTIOC3B, MTIOC3D, MTIOC4A, MTIOC4B, MTIOC4C, and MTIOC4D) in the high-impedance state.

[Setting condition]

- By writing 1 to the MTUCH34HIZ bit

[Clearing conditions]

- Reset
- By writing 0 to the MTUCH34HIZ bit after reading MTUCH34HIZ = 1

MTUCH67HIZ Bit (MTU6 and MTU7 Pin High-Impedance Enable)

This bit specifies whether to put the outputs of the MTU complementary PWM output pins (MTIOC6B, MTIOC6D, MTIOC7A, MTIOC7B, MTIOC7C, and MTIOC7D) in the high-impedance state.

[Setting condition]

- By writing 1 to the MTUCH67HIZ bit

[Clearing conditions]

- Reset
- By writing 0 to the MTUCH67HIZ bit after reading MTUCH67HIZ = 1

MTUCH0HIZ Bit (MTU0 Pin High-Impedance Enable)

This bit specifies whether to put the outputs of the MTU0 pins in the high-impedance state.

[Setting condition]

- By writing 1 to the MTUCH0HIZ bit

[Clearing conditions]

- Reset
- By writing 0 to the MTUCH0HIZ bit after reading MTUCH0HIZ = 1

25.2.11 Port Output Enable Control Register 1 (POECR1)

Address(es): POE3.POECR1 0008 C4CBh

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	MTU0DZE	MTU0CZE	MTU0BZE	MTU0AZE
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	MTU0AZE	MTIOC0A Pin High-Impedance Enable	0: Does not switch the pin to high-impedance state. 1: Switch the pin to high-impedance state.	R/W*1
b1	MTU0BZE	MTIOC0B Pin High-Impedance Enable	0: Does not switch the pin to high-impedance state. 1: Switch the pin to high-impedance state.	R/W*1
b2	MTU0CZE	MTIOC0C Pin High-Impedance Enable	0: Does not switch the pin to high-impedance state. 1: Switch the pin to high-impedance state.	R/W*1
b3	MTU0DZE	MTIOC0D Pin High-Impedance Enable	0: Does not switch the pin to high-impedance state. 1: Switch the pin to high-impedance state.	R/W*1
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

The POECR1 register controls high-impedance state of the MTU0 pins.

MTU0AZE Bit (MTIOC0A Pin High-Impedance Enable)

This bit specifies whether to switch the MTIOC0A output to the high-impedance state when any of the ICSR3.POE8F flag, SPOER.MTUCH0HIZ bit, and ICSR6.OSTSTF flag (when the OSTSTE bit is 1), or, as additionally specified in the POECR5 register, the ICSRn.POE_mF flag (n = 1, 2, 4, 5; m = 0, 4, 10, 11), is set to 1.

MTU0BZE Bit (MTIOC0B Pin High-Impedance Enable)

This bit specifies whether to switch the MTIOC0B output to the high-impedance state when any of the ICSR3.POE8F flag, SPOER.MTUCH0HIZ bit, and ICSR6.OSTSTF flag (when the OSTSTE bit is 1), or, as additionally specified in the POECR5 register, the ICSRn.POE_mF flag (n = 1, 2, 4, 5; m = 0, 4, 10, 11), is set to 1.

MTU0CZE Bit (MTIOC0C Pin High-Impedance Enable)

This bit specifies whether to switch the MTIOC0C output to the high-impedance state when any of the ICSR3.POE8F flag, SPOER.MTUCH0HIZ bit, and ICSR6.OSTSTF flag (when the OSTSTE bit is 1), or, as additionally specified in the POECR5 register, the ICSRn.POE_mF flag (n = 1, 2, 4, 5; m = 0, 4, 10, 11), is set to 1.

MTU0DZE Bit (MTIOC0D Pin High-Impedance Enable)

This bit specifies whether to switch the MTIOC0D output to the high-impedance state when any of the ICSR3.POE8F flag, SPOER.MTUCH0HIZ bit, and ICSR6.OSTSTF flag (when the OSTSTE bit is 1), or, as additionally specified in the POECR5 register, the ICSRn.POE_mF flag (n = 1, 2, 4, 5; m = 0, 4, 10, 11), is set to 1.

25.2.12 Port Output Enable Control Register 2 (POECR2)

Address(es): POE3.POECR2 0008 C4CCh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	MTU3B DZE	MTU4A CZE	MTU4B DZE	—	—	—	—	—	MTU6B DZE	MTU7A CZE	MTU7B DZE
Value after reset:	0	0	0	0	0	1	1	1	0	0	0	0	0	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b0	MTU7BDZE	MTIOC7B/MTIOC7D Pin High-Impedance Enable*2	0: Does not switch the pins to high-impedance state. 1: Switch the pins to high-impedance state.	R/W*1
b1	MTU7ACZE	MTIOC7A/MTIOC7C Pin High-Impedance Enable*2	0: Does not switch the pins to high-impedance state. 1: Switch the pins to high-impedance state.	R/W*1
b2	MTU6BDZE	MTIOC6B/MTIOC6D Pin High-Impedance Enable*2	0: Does not switch the pins to high-impedance state. 1: Switch the pins to high-impedance state.	R/W*1
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	MTU4BDZE	MTIOC4B/MTIOC4D Pin High-Impedance Enable	0: Does not switch the pins to high-impedance state. 1: Switch the pins to high-impedance state.	R/W*1
b9	MTU4ACZE	MTIOC4A/MTIOC4C Pin High-Impedance Enable	0: Does not switch the pins to high-impedance state. 1: Switch the pins to high-impedance state.	R/W*1
b10	MTU3BDZE	MTIOC3B/MTIOC3D Pin High-Impedance Enable	0: Does not switch the pins to high-impedance state. 1: Switch the pins to high-impedance state.	R/W*1
b15 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

Note 2. Set this bit to 0 when MTU6 or MTU7 is not used.

The POECR2 register controls high-impedance state of the MTU complementary PWM output pins (MTU3, MTU4, MTU6, and MTU7 pins).

MTU7BDZE Bit (MTIOC7B/MTIOC7D Pin High-Impedance Enable)

This bit specifies whether to switch the MTIOC7B output and MTIOC7D output to the high-impedance state when at least one of the OCSR2.OSF2 flag, ICSR2.POE4F flag, SPOER.MTUCH67HIZ bit, ICSR6.OSTSTF flag (when the OSTSTE bit is 1), or, as additionally specified in the POECR4 register, the ICSRn.POE_mF flag (n = 1, 3 to 5; m = 0, 8, 10, 11), is set to 1.

MTU7ACZE Bit (MTIOC7A/MTIOC7C Pin High-Impedance Enable)

This bit specifies whether to switch the MTIOC7A output and MTIOC7C output to the high-impedance state when at least one of the OCSR2.OSF2 flag, ICSR2.POE4F flag, SPOER.MTUCH67HIZ bit, ICSR6.OSTSTF flag (when the OSTSTE bit is 1), or, as additionally specified in the POECR4 register, the ICSRn.POE_mF flag (n = 1, 3 to 5; m = 0, 8, 10, 11), is set to 1.

MTU6BDZE Bit (MTIOC6B/MTIOC6D Pin High-Impedance Enable)

This bit specifies whether to switch the MTIOC6B output and MTIOC6D output to the high-impedance state when at least one of the OCSR2.OSF2 flag, ICSR2.POE4F flag, SPOER.MTUCH67HIZ bit, ICSR6.OSTSTF flag (when the OSTSTE bit is 1), or, as additionally specified in the POECR4 register, the ICSRn.POE_mF flag (n = 1, 3 to 5; m = 0, 8, 10, 11), is set to 1.

MTU4BDZE Bit (MTIOC4B/MTIOC4D Pin High-Impedance Enable)

This bit specifies whether to switch the MTIOC4B output and MTIOC4D output to the high-impedance state when at least one of the OCSR1.OSF1 flag, ICSR1.POE0F flag, SPOER.MTUCH34HIZ bit, ICSR6.OSTSTF flag (when the

OSTSTE bit is 1), or, as additionally specified in the POECR4 register, the ICSRn.POE_mF flag (n = 2 to 5; m = 4, 8, 10, 11), is set to 1.

MTU4ACZE Bit (MTIOC4A/MTIOC4C Pin High-Impedance Enable)

This bit specifies whether to switch the MTIOC4A output and MTIOC4C output to the high-impedance state when at least one of the OCSR1.OSF1 flag, ICSR1.POE0F flag, SPOER.MTUCH34HIZ bit, ICSR6.OSTSTF flag (when the OSTSTE bit is 1), or, as additionally specified in the POECR4 register, the ICSRn.POE_mF flag (n = 2 to 5; m = 4, 8, 10, 11), is set to 1.

MTU3BDZE Bit (MTIOC3B/MTIOC3D Pin High-Impedance Enable)

This bit specifies whether to switch the MTIOC3B output and MTIOC3D output to the high-impedance state when at least one of the OCSR1.OSF1 flag, ICSR1.POE0F flag, SPOER.MTUCH34HIZ bit, ICSR6.OSTSTF flag (when the OSTSTE bit is 1), or, as additionally specified in the POECR4 register, the ICSRn.POE_mF flag (n = 2 to 5; m = 4, 8, 10, 11), is set to 1.

25.2.13 Port Output Enable Control Register 4 (POECR4)

Address(es): POE3.POECR4 0008 C4D0h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	IC5ADD MT67ZE	IC4ADD MT67ZE	IC3ADD MT67ZE	—	IC1ADD MT67ZE	—	—	—	IC5ADD MT34ZE	IC4ADD MT34ZE	IC3ADD MT34ZE	IC2ADD MT34ZE	—	—
Value after reset:	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b1	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b2	IC2ADDMT34ZE	MTU3 and MTU4 High-Impedance Condition POE4F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b3	IC3ADDMT34ZE	MTU3 and MTU4 High-Impedance Condition POE8F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b4	IC4ADDMT34ZE	MTU3 and MTU4 High-Impedance Condition POE10F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b5	IC5ADDMT34ZE	MTU3 and MTU4 High-Impedance Condition POE11F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b8 to b6	—	Reserved	These bits are read as 0. The write value should be 0	R/W
b9	IC1ADDMT67ZE	MTU6 and MTU7 High-Impedance Condition POE0F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b10	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b11	IC3ADDMT67ZE	MTU6 and MTU7 High-Impedance Condition POE8F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b12	IC4ADDMT67ZE	MTU6 and MTU7 High-Impedance Condition POE10F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b13	IC5ADDMT67ZE	MTU6 and MTU7 High-Impedance Condition POE11F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

The POECR4 register is used to extend the control conditions to put the output of the MTU complementary PWM output pins (MTU3, MTU4, MTU6, and MTU7) in the high-impedance state.

IC2ADDMT34ZE Bit (MTU3 and MTU4 High-Impedance Condition POE4F Add)

Adds the ICSR2.POE4F flag to the high-impedance control conditions for the MTU3 and MTU4 pins (MTIOC3B, MTIOC3D, MTIOC4A, MTIOC4C, MTIOC4B, MTIOC4D).

IC3ADDMT34ZE Bit (MTU3 and MTU4 High-Impedance Condition POE8F Add)

Adds the ICSR3.POE8F flag to the high-impedance control conditions for the MTU3 and MTU4 pins (MTIOC3B, MTIOC3D, MTIOC4A, MTIOC4C, MTIOC4B, MTIOC4D).

IC4ADDMT34ZE Bit (MTU3 and MTU4 High-Impedance Condition POE10F Add)

Adds the ICSR4.POE10F flag to the high-impedance control conditions for the MTU3 and MTU4 pins (MTIOC3B, MTIOC3D, MTIOC4A, MTIOC4C, MTIOC4B, MTIOC4D).

IC5ADDMT34ZE Bit (MTU3 and MTU4 High-Impedance Condition POE11F Add)

Adds the ICSR5.POE11F flag to the high-impedance control conditions for the MTU3 and MTU4 pins (MTIOC3B, MTIOC3D, MTIOC4A, MTIOC4C, MTIOC4B, MTIOC4D).

IC1ADDMT67ZE Bit (MTU6 and MTU7 High-Impedance Condition POE0F Add)

Adds the ICSR1.POE0F flag to the high-impedance control conditions for the MTU6 and MTU7 pins (MTIOC6B, MTIOC6D, MTIOC7A, MTIOC7C, MTIOC7B, MTIOC7D).

IC3ADDMT67ZE Bit (MTU6 and MTU7 High-Impedance Condition POE8F Add)

Adds the ICSR3.POE8F flag to the high-impedance control conditions for the MTU6 and MTU7 pins (MTIOC6B, MTIOC6D, MTIOC7A, MTIOC7C, MTIOC7B, MTIOC7D).

IC4ADDMT67ZE Bit (MTU6 and MTU7 High-Impedance Condition POE10F Add)

Adds the ICSR4.POE10F flag to the high-impedance control conditions for the MTU6 and MTU7 pins (MTIOC6B, MTIOC6D, MTIOC7A, MTIOC7C, MTIOC7B, MTIOC7D).

IC5ADDMT67ZE Bit (MTU6 and MTU7 High-Impedance Condition POE11F Add)

Adds the ICSR5.POE11F flag to the high-impedance control conditions for the MTU6 and MTU7 pins (MTIOC6B, MTIOC6D, MTIOC7A, MTIOC7C, MTIOC7B, MTIOC7D).

25.2.14 Port Output Enable Control Register 5 (POECR5)

Address(es): POE3.POECR5 0008 C4D2h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	IC5ADD MT0ZE	IC4ADD MT0ZE	—	IC2ADD MT0ZE	IC1ADD MT0ZE	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b1	IC1ADDMT0ZE	MTU0 High-Impedance Condition POE0F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b2	IC2ADDMT0ZE	MTU0 High-Impedance Condition POE4F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b3	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b4	IC4ADDMT0ZE	MTU0 High-Impedance Condition POE10F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b5	IC5ADDMT0ZE	MTU0 High-Impedance Condition POE11F Add	0: Does not add the flag to the conditions to put the output in the high-impedance state. 1: Adds the flag to the conditions to put the output in the high-impedance state.	R/W*1
b15 to b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Can be modified only once after a reset.

The POECR5 register is used to extend the control conditions to put the output of the MTU0 pins in the high-impedance state.

IC1ADDMT0ZE Bit (MTU0 High-Impedance Condition POE0F Add)

Adds the ICSR1.POE0F flag to the high-impedance control conditions for the MTU0 pin (MTIOC0A, MTIOC0B, MTIOC0C, MTIOC0D).

IC2ADDMT0ZE Bit (MTU0 High-Impedance Condition POE4F Add)

Adds the ICSR2.POE4F flag to the high-impedance control conditions for the MTU0 pin (MTIOC0A, MTIOC0B, MTIOC0C, MTIOC0D).

IC4ADDMT0ZE Bit (MTU0 High-Impedance Condition POE10F Add)

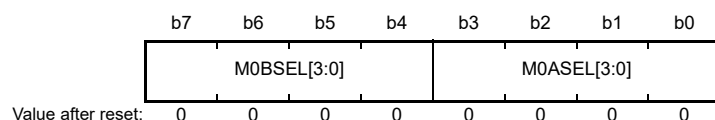
Adds the ICSR4.POE10F flag to the high-impedance control conditions for the MTU0 pin (MTIOC0A, MTIOC0B, MTIOC0C, MTIOC0D).

IC5ADDMT0ZE Bit (MTU0 High-Impedance Condition POE11F Add)

Adds the ICSR5.POE11F flag to the high-impedance control conditions for the MTU0 pin (MTIOC0A, MTIOC0B, MTIOC0C, MTIOC0D).

25.2.15 MTU0 Pin Select Register 1 (M0SELR1)

Address(es): POE3.M0SELR1 0008 C4E4h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	M0ASEL[3:0]	MTU0-A (MTIOC0A) Pin Select	b3 b0 0000: Controls the high-impedance state of P34 on the assumption that the MTIOC0A pin is assigned to P34. 0010: Controls the high-impedance state of PB3 on the assumption that the MTIOC0A pin is assigned to PB3. Settings other than above are prohibited.	R/W*1
b7 to b4	M0BSEL[3:0]	MTU0-B (MTIOC0B) Pin Select	b7 b4 0000: Controls the high-impedance state of P13 on the assumption that the MTIOC0B pin is assigned to P13. 0001: Controls the high-impedance state of P15 on the assumption that the MTIOC0B pin is assigned to P15. 0010: Controls the high-impedance state of PA1 on the assumption that the MTIOC0B pin is assigned to PA1. Settings other than above are prohibited.	R/W*1

Note 1. Can be modified only once after a reset.

The M0SELR1 register is an 8-bit readable/writable register that selects the MTU0-A/B pins as targets for high-impedance control.

M0ASEL[3:0] Bits (MTU0-A (MTIOC0A) Pin Select)

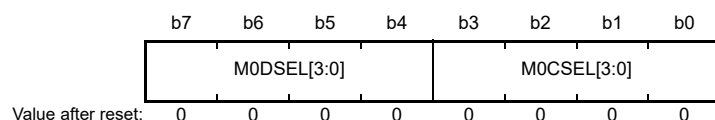
These bits select the target MTIOC0A pin for high-impedance control.

M0BSEL[3:0] Bits (MTU0-B (MTIOC0B) Pin Select)

These bits select the target MTIOC0B pin for high-impedance control.

25.2.16 MTU0 Pin Select Register 2 (M0SELR2)

Address(es): POE3.M0SELR2 0008 C4E5h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	M0CSEL[3:0]	MTU0-C (MTIOC0C) Pin Select	b3 b0 0000: Controls the high-impedance state of P32 on the assumption that the MTIOC0C pin is assigned to P32. 0010: Controls the high-impedance state of PB1 on the assumption that the MTIOC0C pin is assigned to PB1. Settings other than above are prohibited.	R/W*1
b7 to b4	M0DSEL[3:0]	MTU0-D (MTIOC0D) Pin Select	b7 b4 0000: Controls the high-impedance state of P33 on the assumption that the MTIOC0D pin is assigned to P33. 0010: Controls the high-impedance state of PA3 on the assumption that the MTIOC0D pin is assigned to PA3. Settings other than above are prohibited.	R/W*1

Note 1. Can be modified only once after a reset.

The M0SELR2 register is an 8-bit readable/writable register that selects the MTU0-C/D pins as targets for high-impedance control.

M0CSEL[3:0] Bits (MTU0-C (MTIOC0C) Pin Select)

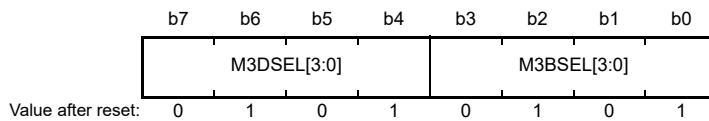
These bits select the target MTIOC0C pin for high-impedance control.

M0DSEL[3:0] Bits (MTU0-D (MTIOC0D) Pin Select)

These bits select the target MTIOC0D pin for high-impedance control.

25.2.17 MTU3 Pin Select Register (M3SELR)

Address(es): POE3.M3SELR 0008 C4E6h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	M3BSEL[3:0]	MTU3-B (MTIOC3B) Pin Select	b3 b0 0000: Controls the high-impedance state of PE1 on the assumption that the MTIOC3B pin is assigned to PE1. 0001: Controls the high-impedance state of P22 on the assumption that the MTIOC3B pin is assigned to P22. 0010: Controls the high-impedance state of P80 on the assumption that the MTIOC3B pin is assigned to P80.*2 0011: Controls the high-impedance state of PC5 on the assumption that the MTIOC3B pin is assigned to PC5. 0100: Controls the high-impedance state of PB7 on the assumption that the MTIOC3B pin is assigned to PB7. 0101: Controls the high-impedance state of P17 on the assumption that the MTIOC3B pin is assigned to P17. Settings other than above are prohibited.	R/W*1
b7 to b4	M3DSEL[3:0]	MTU3-D (MTIOC3D) Pin Select	b7 b4 0000: Controls the high-impedance state of PE0 on the assumption that the MTIOC3D pin is assigned to PE0. 0001: Controls the high-impedance state of P23 on the assumption that the MTIOC3D pin is assigned to P23. 0010: Controls the high-impedance state of PC4 on the assumption that the MTIOC3D pin is assigned to PC4. 0011: Controls the high-impedance state of P81 on the assumption that the MTIOC3D pin is assigned to P81.*2 0100: Controls the high-impedance state of PB6 on the assumption that the MTIOC3D pin is assigned to PB6. 0101: Controls the high-impedance state of P16 on the assumption that the MTIOC3D pin is assigned to P16. Settings other than above are prohibited.	R/W*1

Note 1. Can be modified only once after a reset.

Note 2. This selection is only usable in products with at least 144 pins.

The M3SELR register is an 8-bit readable/writable register that selects the MTU3-B/D pins as targets for high-impedance control.

M3BSEL[3:0] Bits (MTU3-B (MTIOC3B) Pin Select)

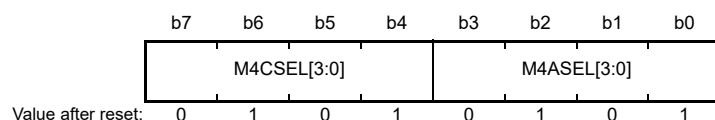
These bits select the target MTIOC3B pin for high-impedance control.

M3DSEL[3:0] Bits (MTU3-D (MTIOC3D) Pin Select)

These bits select the target MTIOC3D pin for high-impedance control.

25.2.18 MTU4 Pin Select Register 1 (M4SELR1)

Address(es): POE3.M4SELR1 0008 C4E7h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	M4ASEL[3:0]	MTU4-A (MTIOC4A) Pin Select	b3 b0 0000: Controls the high-impedance state of PE2 on the assumption that the MTIOC4A pin is assigned to PE2. 0001: Controls the high-impedance state of P21 on the assumption that the MTIOC4A pin is assigned to P21. 0010: Controls the high-impedance state of PB3 on the assumption that the MTIOC4A pin is assigned to PB3. 0011: Controls the high-impedance state of P82 on the assumption that the MTIOC4A pin is assigned to P82.*2 0100: Controls the high-impedance state of PA0 on the assumption that the MTIOC4A pin is assigned to PA0. 0101: Controls the high-impedance state of P24 on the assumption that the MTIOC4A pin is assigned to P24. Settings other than above are prohibited.	R/W*1
b7 to b4	M4CSEL[3:0]	MTU4-C (MTIOC4C) Pin Select	b7 b4 0000: Controls the high-impedance state of PE5 on the assumption that the MTIOC4C pin is assigned to PE5. 0001: Controls the high-impedance state of P87 on the assumption that the MTIOC4C pin is assigned to P87.*2 0010: Controls the high-impedance state of PB1 on the assumption that the MTIOC4C pin is assigned to PB1. 0011: Controls the high-impedance state of P83 on the assumption that the MTIOC4C pin is assigned to P83.*2 0100: Controls the high-impedance state of PE1 on the assumption that the MTIOC4C pin is assigned to PE1. 0101: Controls the high-impedance state of P25 on the assumption that the MTIOC4C pin is assigned to P25. Settings other than above are prohibited.	R/W*1

Note 1. Can be modified only once after a reset.

Note 2. This selection is only usable in products with at least 144 pins.

The M4SELR1 register is an 8-bit readable/writable register that selects the MTU4-A/C pins as targets for high-impedance control.

M4ASEL[3:0] Bits (MTU4-A (MTIOC4A) Pin Select)

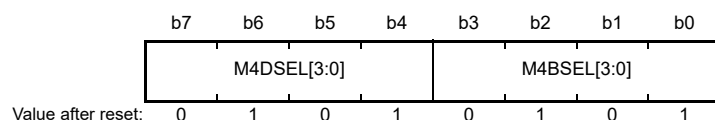
These bits select the target MTIOC4A pin for high-impedance control.

M4CSEL[3:0] Bits (MTU4-C (MTIOC4C) Pin Select)

These bits select the target MTIOC4C pin for high-impedance control.

25.2.19 MTU4 Pin Select Register 2 (M4SELR2)

Address(es): POE3.M4SELR2 0008 C4E8h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	M4BSEL[3:0]	MTU4-B (MTIOC4B) Pin Select	b3 b0 0000: Controls the high-impedance state of PE3 on the assumption that the MTIOC4B pin is assigned to PE3. 0001: Controls the high-impedance state of P17 on the assumption that the MTIOC4B pin is assigned to P17. 0010: Controls the high-impedance state of P54 on the assumption that the MTIOC4B pin is assigned to P54. 0011: Controls the high-impedance state of PC2 on the assumption that the MTIOC4B pin is assigned to PC2. 0100: Controls the high-impedance state of PD1 on the assumption that the MTIOC4B pin is assigned to PD1. 0101: Controls the high-impedance state of P30 on the assumption that the MTIOC4B pin is assigned to P30. Settings other than above are prohibited.	R/W*1
b7 to b4	M4DSEL[3:0]	MTU4-D (MTIOC4D) Pin Select	b7 b4 0000: Controls the high-impedance state of PE4 on the assumption that the MTIOC4D pin is assigned to PE4. 0001: Controls the high-impedance state of P86 on the assumption that the MTIOC4D pin is assigned to P86.*2 0010: Controls the high-impedance state of P55 on the assumption that the MTIOC4D pin is assigned to P55. 0011: Controls the high-impedance state of PC3 on the assumption that the MTIOC4D pin is assigned to PC3. 0100: Controls the high-impedance state of PD2 on the assumption that the MTIOC4D pin is assigned to PD2. 0101: Controls the high-impedance state of P31 on the assumption that the MTIOC4D pin is assigned to P31. Settings other than above are prohibited.	R/W*1

Note 1. Can be modified only once after a reset.

Note 2. This selection is only usable in products with at least 144 pins.

The M4SELR2 register is an 8-bit readable/writable register that selects the MTU4-B/D pins as targets for high-impedance control.

M4BSEL[3:0] Bits (MTU4-B (MTIOC4B) Pin Select)

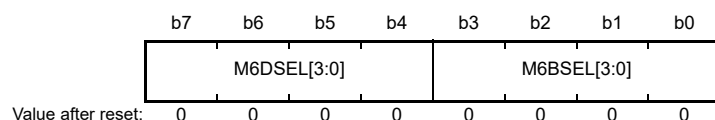
These bits select the target MTIOC4A pin for high-impedance control.

M4DSEL[3:0] Bits (MTU4-D (MTIOC4D) Pin Select)

These bits select the target MTIOC4C pin for high-impedance control.

25.2.20 MTU6 Pin Select Register (M6SELR)

Address(es): POE3.M6SELR 0008 C4EAh



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	M6BSEL[3:0]	MTU6-B (MTIOC6B) Pin Select*2	b3 b0 0000: Controls the high-impedance state of PA5 on the assumption that the MTIOC6B pin is assigned to PA5. 0001: Controls the high-impedance state of PJ0 on the assumption that the MTIOC6B pin is assigned to PJ0.*3 Settings other than above are prohibited.	R/W*1
b7 to b4	M6DSEL[3:0]	MTU6-D (MTIOC6D) Pin Select*2	b7 b4 0000: Controls the high-impedance state of PA0 on the assumption that the MTIOC6D pin is assigned to PA0. 0001: Controls the high-impedance state of P84 on the assumption that the MTIOC6D pin is assigned to P84.*3 Settings other than above are prohibited.	R/W*1

Note 1. Can be modified only once after a reset.

Note 2. These bits are only effective in products with at least 176 pins. Set these bits to 0000b in products with fewer pins.

Note 3. This selection is only usable in products with at least 176 pins.

The M6SELR register is an 8-bit readable/writable register that selects the MTU6-B/D pins as targets for high-impedance control.

M6BSEL[3:0] Bits (MTU6-B (MTIOC6B) Pin Select)

These bits select the target MTIOC6B pin for high-impedance control.

M6DSEL[3:0] Bits (MTU6-D (MTIOC6D) Pin Select)

These bits select the target MTIOC6D pin for high-impedance control.

25.3 Operation

The following shows the target pins and conditions for high-impedance control.

(1) MTU3 pins (MTIOC3B, MTIOC3D)

When one of the following conditions is satisfied while the POECR2.MTU3BDZE bit is 1, the pins become high-impedance.

- Operation for detection of the POE0# input level
When the ICSR1.POE0F flag becomes 1.
- Operation for comparison of the output levels on the MTIOC3B and MTIOC3D pins
When the OCSR1.OSF1 flag becomes 1 while the OCSR1.OCE1 bit is 1.
- SPOER setting
When the SPOER.MTUCH34HIZ bit is set to 1.
- Conditions added by POECR4
When the ICSR2.POE4F flag becomes 1 while the POECR4.IC2ADDMT34ZE bit is 1.
When the ICSR3.POE8F flag becomes 1 while the POECR4.IC3ADDMT34ZE bit and the ICSR3.POE8E bit are 1.
When the ICSR4.POE10F flag becomes 1 while the POECR4.IC4ADDMT34ZE bit and the ICSR4.POE10E bit are 1.
When the ICSR5.POE11F flag becomes 1 while the POECR4.IC5ADDMT34ZE bit and the ICSR5.POE11E bit are 1.
- Detection of oscillation stop
When the ICSR6.OSTSTF flag becomes 1 while the ICSR6.OSTSTE bit is 1.

(2) MTU4 pins (MTIOC4A, MTIOC4C)

When one of the following conditions is satisfied while the POECR2.MTU4ACZE bit is 1, the pins become high-impedance.

- Operation for detection of the POE0# input level
When the ICSR1.POE0F flag becomes 1.
- Operation for comparison of the output levels on the MTIOC4A and MTIOC4C pins
When the OCSR1.OSF1 flag becomes 1 while the OCSR1.OCE1 bit is 1.
- SPOER setting
When the SPOER.MTUCH34HIZ bit is set to 1.
- Conditions added by POECR4
When the ICSR2.POE4F flag becomes 1 while the POECR4.IC2ADDMT34ZE bit is 1.
When the ICSR3.POE8F flag becomes 1 while the POECR4.IC3ADDMT34ZE bit and the ICSR3.POE8E bit are 1.
When the ICSR4.POE10F flag becomes 1 while the POECR4.IC4ADDMT34ZE bit and the ICSR4.POE10E bit are 1.
When the ICSR5.POE11F flag becomes 1 while the POECR4.IC5ADDMT34ZE bit and the ICSR5.POE11E bit are 1.
- Detection of oscillation stop
When the ICSR6.OSTSTF flag becomes 1 while the ICSR6.OSTSTE bit is 1.

(3) MTU4 pins (MTIOC4B, MTIOC4D)

When one of the following conditions is satisfied while the POECR2.MTU4BDZE bit is 1, the pins become high-impedance.

- Operation for detection of the POE0# input level
When the ICSR1.POE0F flag becomes 1.
- Operation for comparison of the output levels on the MTIOC4B and MTIOC4D pins

When the OCSR1.OSF1 flag becomes 1 while the OCSR1.OCE1 bit is 1.

- SPOER setting

When the SPOER.MTUCH34HIZ bit is set to 1.

- Conditions added by POECR4

When the ICSR2.POE4F flag becomes 1 while the POECR4.IC2ADDMT34ZE bit is 1.

When the ICSR3.POE8F flag becomes 1 while the POECR4.IC3ADDMT34ZE bit and the ICSR3.POE8E bit are 1.

When the ICSR4.POE10F flag becomes 1 while the POECR4.IC4ADDMT34ZE bit and the ICSR4.POE10E bit are 1.

When the ICSR5.POE11F flag becomes 1 while the POECR4.IC5ADDMT34ZE bit and the ICSR5.POE11E bit are 1.

- Detection of oscillation stop

When the ICSR6.OSTSTF flag becomes 1 while the ICSR6.OSTSTE bit is 1.

(4) MTU6 pins (MTIOC6B, MTIOC6D)

When one of the following conditions is satisfied while the POECR2.MTU6BDZE bit is 1, the pins become high-impedance.

- Operation for detection of the POE4# input level

When the ICSR2.POE4F flag becomes 1.

- Operation for comparison of the output levels on the MTIOC6B and MTIOC6D pins

When the OCSR2.OSF2 flag becomes 1 while the OCSR2.OCE2 bit is 1.

- SPOER setting

When the SPOER.MTUCH67HIZ bit is set to 1.

- Conditions added by POECR4

When the ICSR1.POE0F flag becomes 1 while the POECR4.IC1ADDMT67ZE bit is 1.

When the ICSR3.POE8F flag becomes 1 while the POECR4.IC3ADDMT67ZE bit and the ICSR3.POE8E bit are 1.

When the ICSR4.POE10F flag becomes 1 while the POECR4.IC4ADDMT67ZE bit and the ICSR4.POE10E bit are 1.

When the ICSR5.POE11F flag becomes 1 while the POECR4.IC5ADDMT67ZE bit and the ICSR5.POE11E bit are 1.

- Detection of oscillation stop

When the ICSR6.OSTSTF flag becomes 1 while the ICSR6.OSTSTE bit is 1.

(5) MTU7 pins (MTIOC7A, MTIOC7C)

When one of the following conditions is satisfied while the POECR2.MTU7ACZE bit is 1, the pins become high-impedance.

- Operation for detection of the POE4# input level

When the ICSR2.POE4F flag becomes 1.

- Operation for comparison of the output levels on the MTIOC7A and MTIOC7C pins

When the OCSR2.OSF2 flag becomes 1 while the OCSR2.OCE2 bit is 1.

- SPOER setting

When the SPOER.MTUCH67HIZ bit is set to 1.

- Conditions added by POECR4

When the ICSR1.POE0F flag becomes 1 while the POECR4.IC1ADDMT67ZE bit is 1.

When the ICSR3.POE8F flag becomes 1 while the POECR4.IC3ADDMT67ZE bit and the ICSR3.POE8E bit are 1.

When the ICSR4.POE10F flag becomes 1 while the POECR4.IC4ADDMT67ZE bit and the ICSR4.POE10E bit are 1.

When the ICSR5.POE11F flag becomes 1 while the POECR4.IC5ADDMT67ZE bit and the ICSR5.POE11E bit are 1.

- Detection of oscillation stop
When the ICSR6.OSTSTF flag becomes 1 while the ICSR6.OSTSTE bit is 1.

(6) MTU7 pins (MTIOC7B, MTIOC7D)

When one of the following conditions is satisfied while the POECR2.MTU7BDZE bit is 1, the pins become high-impedance.

- Operation for detection of the POE4# input level
When the ICSR2.POE4F flag becomes 1.
- Operation for comparison of the output levels on the MTIOC7B and MTIOC7D pins
When the OCSR2.OSF2 flag becomes 1 while the OCSR2.OCE2 bit is 1.
- SPOER setting
When the SPOER.MTUCH67HIZ bit is set to 1.
- Conditions added by POECR4
When the ICSR1.POE0F flag becomes 1 while the POECR4.IC1ADDMT67ZE bit is 1.
When the ICSR3.POE8F flag becomes 1 while the POECR4.IC3ADDMT67ZE bit and the ICSR3.POE8E bit are 1.
When the ICSR4.POE10F flag becomes 1 while the POECR4.IC4ADDMT67ZE bit and the ICSR4.POE10E bit are 1.
When the ICSR5.POE11F flag becomes 1 while the POECR4.IC5ADDMT67ZE bit and the ICSR5.POE11E bit are 1.
- Detection of oscillation stop
When the ICSR6.OSTSTF flag becomes 1 while the ICSR6.OSTSTE bit is 1.

(7) MTU0 pin (MTIOC0A)

When one of the following conditions is satisfied while the POECR1.MTU0AZE bit is 1, the pin becomes high-impedance.

- Operation for detection of the POE8# input level
When the ICSR3.POE8F flag becomes 1 while the ICSR3.POE8E bit is 1.
- SPOER setting
When the SPOER.MTUCH0HIZ bit is set to 1.
- Conditions added by POECR5
When the ICSR1.POE0F flag becomes 1 while the POECR5.IC1ADDMT0ZE bit is 1.
When the ICSR2.POE4F flag becomes 1 while the POECR5.IC2ADDMT0ZE bit is 1.
When the ICSR4.POE10F flag becomes 1 while the POECR5.IC4ADDMT0ZE bit and the ICSR4.POE10E bit are 1.
When the ICSR5.POE11F flag becomes 1 while the POECR5.IC5ADDMT0ZE bit and the ICSR5.POE11E bit are 1.
- Detection of oscillation stop
When the ICSR6.OSTSTF flag becomes 1 while the ICSR6.OSTSTE bit is 1.

(8) MTU0 pin (MTIOC0B)

When one of the following conditions is satisfied while the POECR1.MTU0BZE bit is 1, the pin becomes high-impedance.

- Operation for detection of the POE8# input level
When the ICSR3.POE8F flag becomes 1 while the ICSR3.POE8E bit is 1.
- SPOER setting
When the SPOER.MTUCH0HIZ bit is set to 1.
- Conditions added by POECR5
When the ICSR1.POE0F flag becomes 1 while the POECR5.IC1ADDMT0ZE bit is 1.

When the ICSR2.POE4F flag becomes 1 while the POECR5.IC2ADDMT0ZE bit is 1.

When the ICSR4.POE10F flag becomes 1 while the POECR5.IC4ADDMT0ZE bit and the ICSR4.POE10E bit are 1.

When the ICSR5.POE11F flag becomes 1 while the POECR5.IC5ADDMT0ZE bit and the ICSR5.POE11E bit are 1.

- Detection of oscillation stop

When the ICSR6.OSTSTF flag becomes 1 while the ICSR6.OSTSTE bit is 1.

(9) MTU0 pin (MTIOC0C)

When one of the following conditions is satisfied while the POECR1.MTU0CZE bit is 1, the pin becomes high-impedance.

- Operation for detection of the POE8# input level

When the ICSR3.POE8F flag becomes 1 while the ICSR3.POE8E bit is 1.

- SPOER setting

When the SPOER.MTUCH0HIZ bit is set to 1.

- Conditions added by POECR5

When the ICSR1.POE0F flag becomes 1 while the POECR5.IC1ADDMT0ZE bit is 1.

When the ICSR2.POE4F flag becomes 1 while the POECR5.IC2ADDMT0ZE bit is 1.

When the ICSR4.POE10F flag becomes 1 while the POECR5.IC4ADDMT0ZE bit and the ICSR4.POE10E bit are 1.

When the ICSR5.POE11F flag becomes 1 while the POECR5.IC5ADDMT0ZE bit and the ICSR5.POE11E bit are 1.

- Detection of oscillation stop

When the ICSR6.OSTSTF flag becomes 1 while the ICSR6.OSTSTE bit is 1.

(10) MTU0 pin (MTIOC0D)

When one of the following conditions is satisfied while the POECR1.MTU0DZE bit is 1, the pin becomes high-impedance.

- Operation for detection of the POE8# input level

When the ICSR3.POE8F flag becomes 1 while the ICSR3.POE8E bit is 1.

- SPOER setting

When the SPOER.MTUCH0HIZ bit is set to 1.

- Conditions added by POECR5

When the ICSR1.POE0F flag becomes 1 while the POECR5.IC1ADDMT0ZE bit is 1.

When the ICSR2.POE4F flag becomes 1 while the POECR5.IC2ADDMT0ZE bit is 1.

When the ICSR4.POE10F flag becomes 1 while the POECR5.IC4ADDMT0ZE bit and the ICSR4.POE10E bit are 1.

When the ICSR5.POE11F flag becomes 1 while the POECR5.IC5ADDMT0ZE bit and the ICSR5.POE11E bit are 1.

- Detection of oscillation stop

When the ICSR6.OSTSTF flag becomes 1 while the ICSR6.OSTSTE bit is 1.

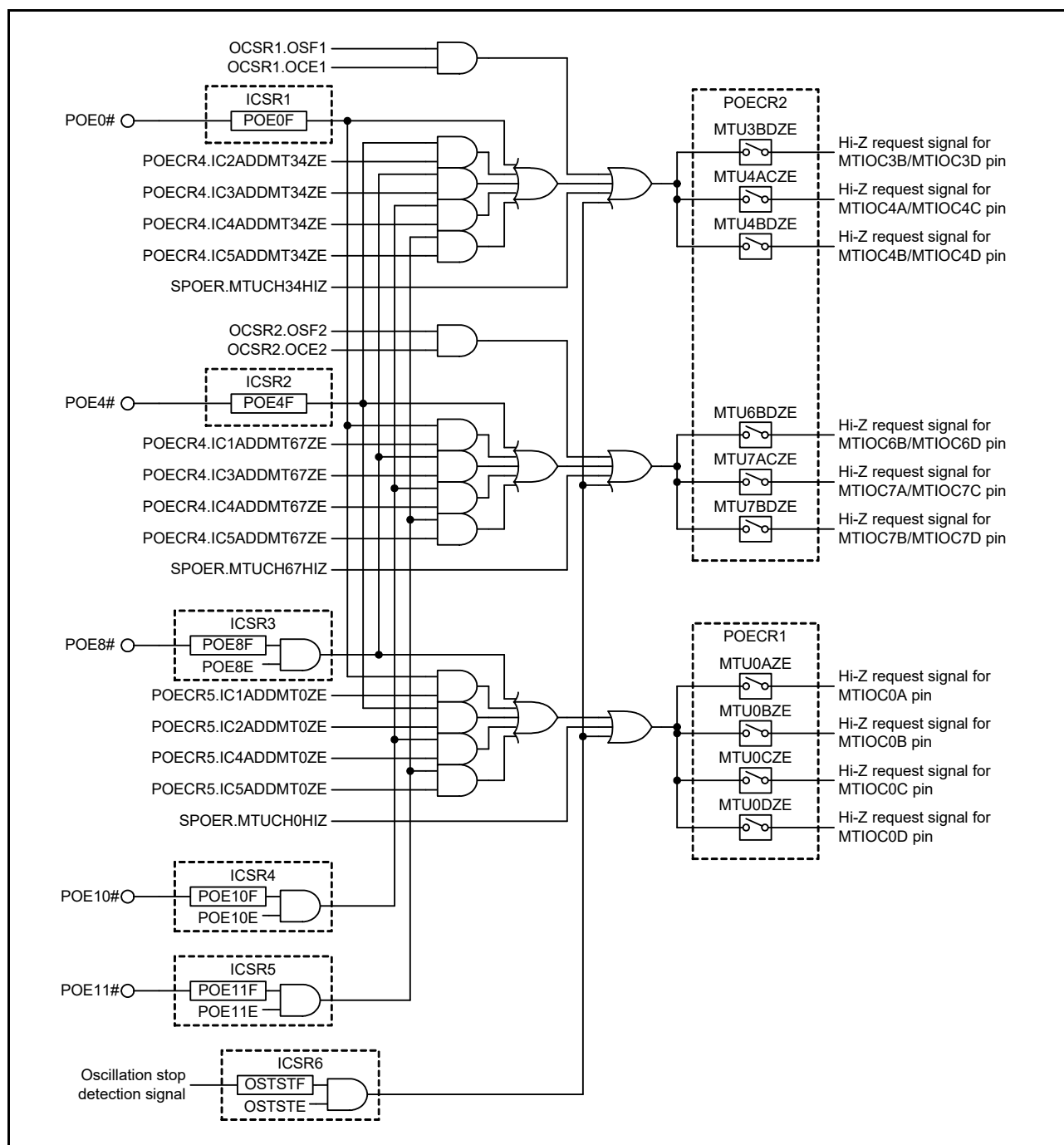


Figure 25.2 Target Pins and Conditions for High-Impedance Control

25.3.1 MTU Pin Selection

In this MCU, the pin functions for MTU are respectively multiplexed with multiple sets of port pins. The target pins for high-impedance control can be selected by the pin select register in POE3 (M0SELR1, M0SELR2, M3SELR, M4SELR1, M4SELR2, or M6SELR register). Table 25.4 shows the correspondence between MTU pins and select registers.

Note that settings for pins to be used as MTU must be separately made in the registers of the multi-function pin controller (MPC). Take care so that there are no differences between the pins selected in the POE3 registers and the pins selected in the MPC registers.

Table 25.4 Correspondence between MTU Pins

MTU Pin Functions	Corresponding Ports	Select Registers
MTIOC0A	P34	M0SELR1
	PB3	
MTIOC0B	P13	
	P15	
	PA1	
MTIOC0C	P32	M0SELR2
	PB1	
MTIOC0D	P33	
	PA3	
MTIOC3B	PE1	M3SELR
	P22	
	P80*1	
	PC5	
	PB7	
	P17	
MTIOC3D	PE0	
	P23	
	PC4	
	P81*1	
	PB6	
	P16	
MTIOC4A	PE2	M4SELR1
	P21	
	PB3	
	P82*1	
	PA0	
	P24	
MTIOC4C	PE5	
	P87*1	
	PB1	
	P83*1	
	PE1	
	P25	

MTU Pin Functions	Corresponding Ports	Select Registers
MTIOC4B	PE3	M4SELR2
	P17	
	P54	
	PC2	
	PD1	
	P30	
MTIOC4D	PE4	M4SELR2
	P86*1	
	P55	
	PC3	
	PD2	
	P31	
MTIOC6B	PA5	M6SELR
	PJ0*2	
MTIOC6D	PA0	M6SELR
	P84*2	
MTIOC7A	PA2	—
MTIOC7C*1	P67*1	
MTIOC7B	PA1	
MTIOC7D*1	P66*1	

Note 1. This is only present in products with at least 144 pins.
Note 2. This is only present in products with at least 176 pins.

Note 1. This is only present in products with at least 144 pins.

Note 2. This is only present in products with at least 176 pins.

25.3.2 Input-Level Detection Operation

If the input conditions set by ICSR1 to ICSR5 occur on the POE0#, POE4#, POE8#, POE10#, and POE11# pins, the outputs of the MTU complementary PWM output pins (MTU3 and MTU4 or MTU6 and MTU7) and MTU0 pins are in the high-impedance state. Note however, that these outputs are still in the high-impedance state even when the MTU functions are not selected for the pins.

(1) Falling Edge Detection

When a change from a high to low level is input to the POE0#, POE4#, POE8#, POE10#, and POE11# pins, the outputs of the pins multiplexed with MTU complementary PWM output pins and MTU0 pins are in the high-impedance state. The falling edge is detected after the level is sampled with PCLK. Input a low level for at least one PCLK clock to the POE0#, POE4#, POE8#, POE10#, and POE11# pins.

Figure 25.3 shows a sample timing after the level changes in input to the POE0#, POE4#, POE8#, POE10#, and POE11# pins until the respective pins become high-impedance.

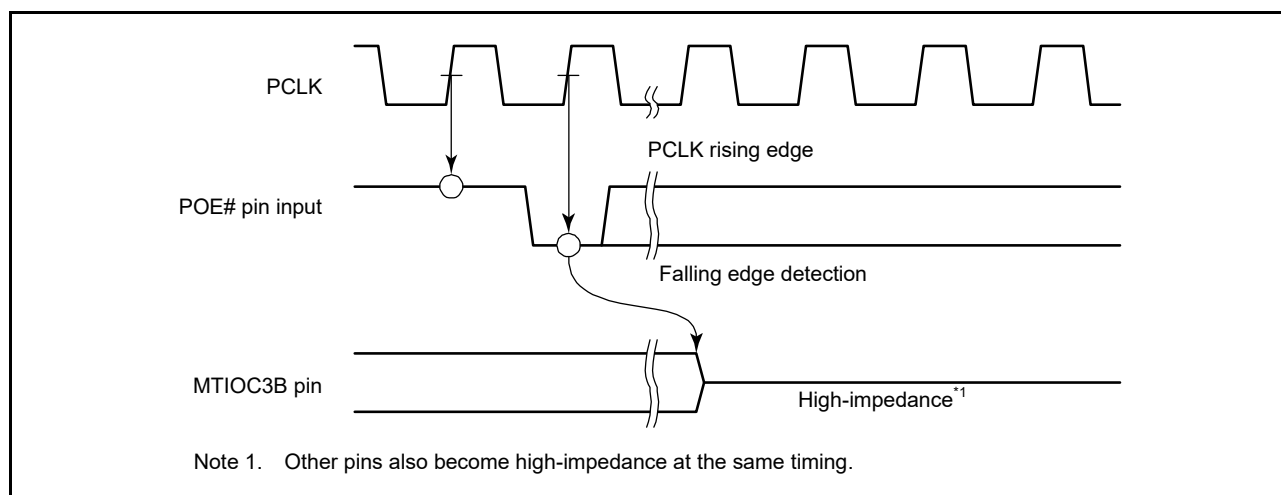


Figure 25.3 Operation when A Falling Edge Detection is Selected

(2) Low-Level Detection

Figure 25.4 shows an example of operation when a pin is placed in the high-impedance state in response to low-level detection. When 16 continuous low levels are sampled with the sampling clock selected by the ICSR1 to ICSR5 registers, the low level is recognized and the outputs of the MTU complementary PWM output pins and MTU0 pins are in the high-impedance state. If even one high level is detected during this interval, the low level is not recognized. The timing when the outputs of the MTU complementary PWM output pins and MTU0 pins are in the high-impedance state after the sampling clock is input is the same in both falling-edge detection and in low-level detection.

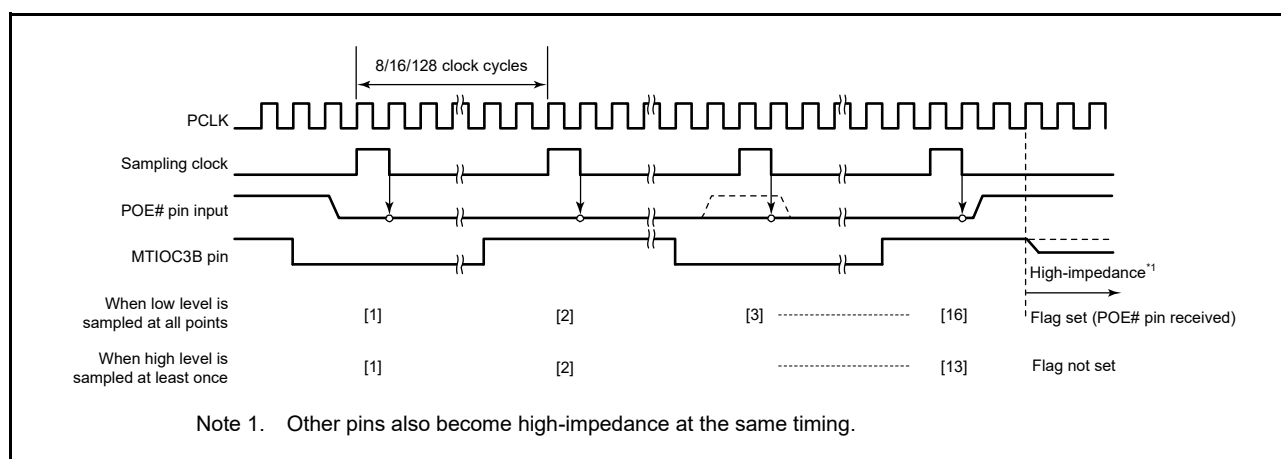


Figure 25.4 Operation when A Low-Level Detection is Selected

25.3.3 Output-Level Compare Operation

Figure 25.5 shows an example of the output-level compare operation for the combination of MTIOC3B and MTIOC3D. The operation is the same for the other pin combinations.

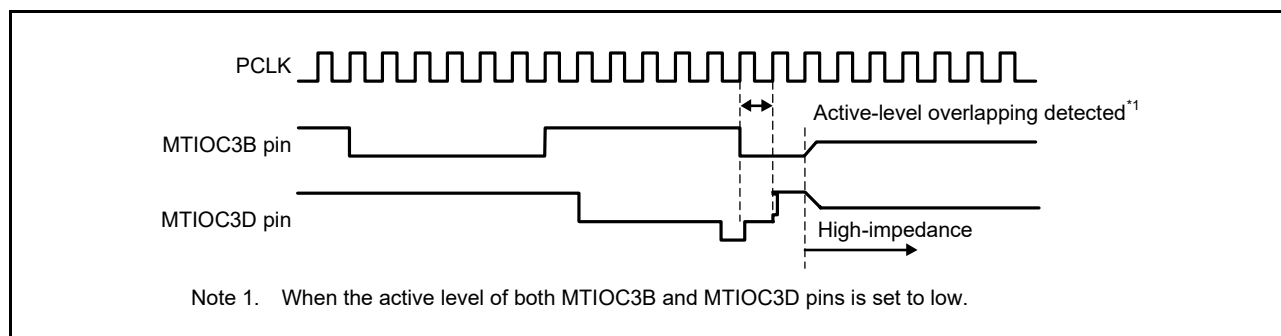


Figure 25.5 Output-Level Compare Operation

25.3.4 High-Impedance Control Using Registers

The high-impedance request of the MTU pins (MTU0, MTU3, MTU4, MTU6, and MTU7) can be directly controlled by using the SPOER register.

For instance, setting the SPOER.MTUCH34HIZ bit to 1 switches the MTU3 and MTU4 pins specified by the POECR2 register to the high-impedance state.

The high-impedance request of other pins can also be controlled by setting the appropriate bits in the SPOER register.

25.3.5 High-Impedance Control through Detection of Oscillation Stop

When oscillation stop is detected by the oscillation stop detection function of the clock generator while the ICSR6.OSTSTE bit is 1, the MTU complementary PWM output pins specified by the POECR2 register and the MTU0 pins specified by the POECR1 register are switched to the high-impedance state.

25.3.6 Additional Functions for High-Impedance Control

High-impedance control conditions for the MTU complementary PWM output pins and MTU0 pins can be added by setting the POECR4 and POECR5 registers.

For instance, the settings listed below can be added as high-impedance control conditions for the MTU3 and MTU4 pins.

- Setting the POECR4.IC2ADDMT34ZE bit to 1 adds the input-level detection by the POE4# pin
- Setting the POECR4.IC3ADDMT34ZE bit to 1 and adds the input-level detection by the POE8# pin
- Setting the POECR4.IC4ADDMT34ZE bit to 1 and adds the input-level detection by the POE10# pin
- Setting the POECR4.IC5ADDMT34ZE bit to 1 and adds the input-level detection by the POE11# pin

The high-impedance control of other pins can also be controlled by setting the appropriate bits in the POECR4 and POECR5 registers.

25.3.7 Recover from High-Impedance State

The outputs which have been in the high-impedance state due to input-level detection can be recovered from the state either by returning them to their initial state with a reset, or by clearing all of the ICSR1.POE0F, ICSR2.POE4F, ICSR3.POE8F, ICSR4.POE10F, and ICSR5.POE11F flags. However, note that when low-level sampling is selected with the ICSR1.POE0M[1:0], ICSR2.POE4M[1:0], ICSR3.POE8M[1:0], ICSR4.POE10M[1:0], and ICSR5.POE11M[1:0] bits, just writing 0 to a flag is ignored (the flag is not set to 0); flags can be cleared by writing 0 to it only after a high level is input to the POE0#, POE4#, POE8#, POE10#, and POE11# pins and is detected.

The outputs which have been in the high-impedance state due to output-level detection can be recovered from the state either by returning them to their initial state with a reset, or by setting the OCSR1.OSF1 flag or the OCSR2.OSF2 flag to 0. However, note that just writing 0 to a flag is ignored (the flag is not set to 0); the flags can be cleared by writing 0 to it only after setting the inactive level to be output from the pin. In the MTU, the inactive level (initial output level) can be output by stopping the count operation.

The outputs which have been in the high-impedance state due to oscillation stop detection can be recovered from the state either by returning them to their initial state with a reset or by setting the SYSTEM.OSTDSR.OSTDF flag to 0 to set the ICSR6.OSTSTF flag to 0.

25.4 POE3 Setting Procedure

Figure 25.6 shows the procedure for setting the POE3. It illustrates an example of high-impedance control in response to comparison of the output levels on the MTU3 pins (MTIOC3B/MTIOC3D). In the figure, P22 is used as the MTIOC3B pin and P23 is used as the MTIOC3D pin.

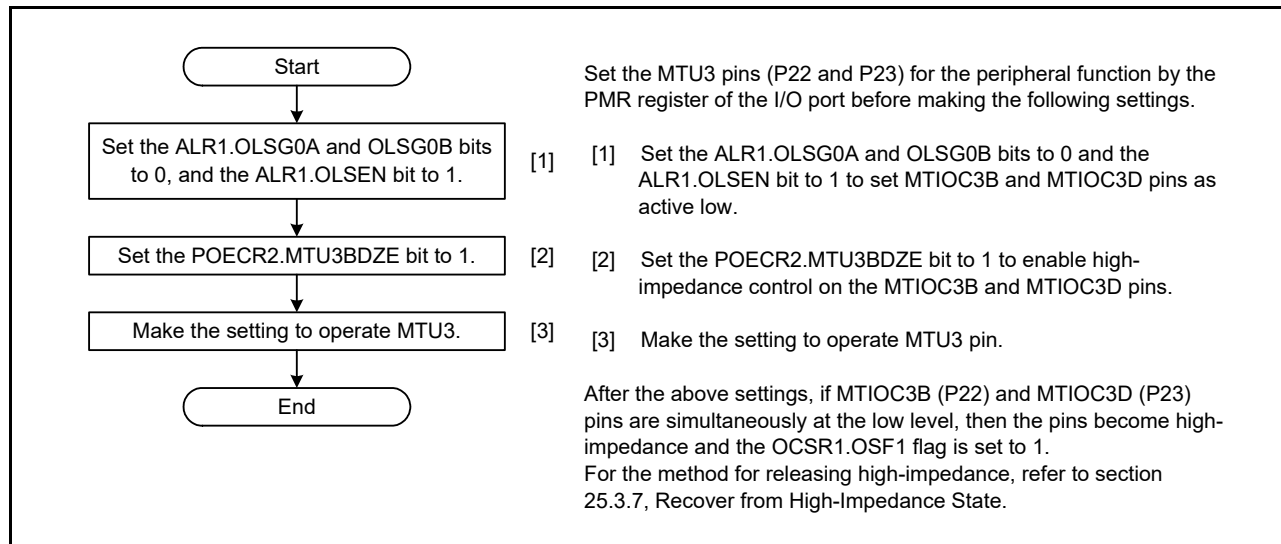


Figure 25.6 Procedure for Setting the POE3

25.5 Interrupts

The POE3 issues a request to generate an interrupt when the specified condition is satisfied during input-level detection or output-level comparison. Table 25.5 shows the interrupt sources and their conditions.

Table 25.5 Interrupt Sources and Conditions

Name	Interrupt Source	Interrupt Flag	Condition
OEI1	Output enable interrupt 1	POE0F OSF1	When the ICSR1.POE0F flag becomes 1 while the ICSR1.PIE1 bit is 1 or when the OCSR1.OSF1 flag becomes 1 while the OCSR1.OIE1 bit is 1
OEI2	Output enable interrupt 2	POE4F OSF2	When the ICSR2.POE4F flag becomes 1 while the ICSR2.PIE2 bit is 1 or when the OCSR2.OSF2 flag becomes 1 while the OCSR2.OIE2 bit is 1
OEI3	Output enable interrupt 3	POE8F	When the ICSR3.POE8F flag becomes 1 while the ICSR3.PIE3 bit is 1
OEI4	Output enable interrupt 4	POE10F POE11F	When the ICSR4.POE10F flag is set to 1 while the ICSR4.PIE4 bit is 1 or when the ICSR5.POE11F flag becomes 1 while the ICSR5.PIE5 bit is 1

25.6 Usage Notes

25.6.1 Transition to Low Power Consumption Mode

When the POE3 is used, do not make a transition to software standby mode or deep software standby mode. In these modes, the POE3 stops and thus the high-impedance control of pins cannot operate.

25.6.2 High-Impedance Control When the MTU is Not Selected

If high-impedance control for a pin having a multiplexed MTU pin function is enabled by setting the POECR1 and POECR2 registers and the high-impedance control condition is satisfied, the output is to be in the high-impedance state even if the MTU function is not selected for the pin on which it is multiplexed.

To avoid unintended high-impedance, ensure that there are no differences between the settings for MTU pin selection in the PmnPFS registers of the MPC and for MTU pin selection in the pin select register of the POE3.

25.6.3 When the POE3 is Not Used

The high-impedance control of some pins can be enabled using the POE3 after a reset. When the POE3 is not used, write 0 to the target bits in the POECR1 and POECR2 registers.